Genetics-informed SuStaIn

- We extend SuStaIn to include a APOE-informed subtype prior probability;
- Categorical variable APOE-status is one-hot encoded;
  Each possible value(0,1,2) has its own weight;
- Derived weight update eq with Lagrange multiplier

$$L = \prod_{j=1}^{M} \left(\sum_{s=1}^{N_S} f_s \cdot (z_{j,0} w_0 + z_{j,1} w_1 + z_{j,2} w_2) \cdot P(X_j | S_s) \right)$$

Where:

- $\mathbf{z}_j$ = **Covariate dummy variables** for patient j
  - $[1,0,0]$ = APOE non-carrier
  - $[0,1,0]$ = APOE monoallelic
  - $[0,0,1]$ = APOE biallelic
- $\mathbf{w}$ = **Phenotypic weights** (w_0, w_1, w_2) for each category

E-step:

$$\gamma_{jks} = \frac{f_s \cdot (z_{j,0} w_0 + z_{j,1} w_1 + z_{j,2} w_2) \cdot p(X_j | S_s, k)}{P_{\text{total}}(j)}$$

M-step:

$$\text{Standard Subtype Frequency:} \quad f_s^{\text{new}} = \frac{\sum_{j=1}^{M} \gamma_{j,s}}{\sum_{s=1}^{N_S} \sum_{j=1}^{M} \gamma_{j,s}}$$

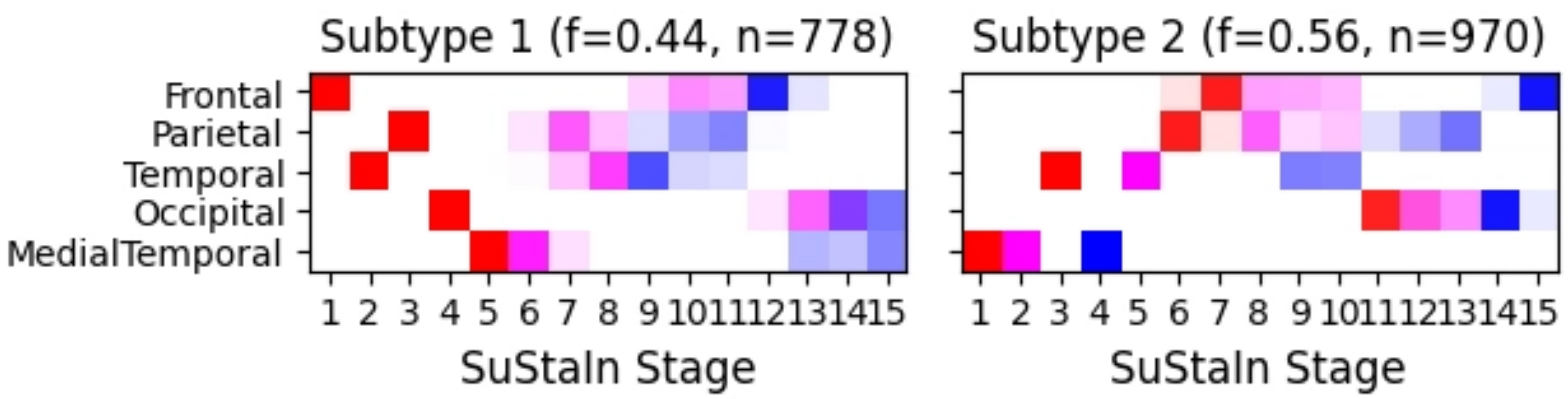
$$\text{Covariate Category Weight:} \quad w_{s,l}^{\text{new}} = \frac{\sum_{j=1}^{M} \gamma_{j,s} \cdot z_{j,l}}{\sum_{s=1}^{N_S} \sum_{j=1}^{M} \gamma_{j,s}}$$

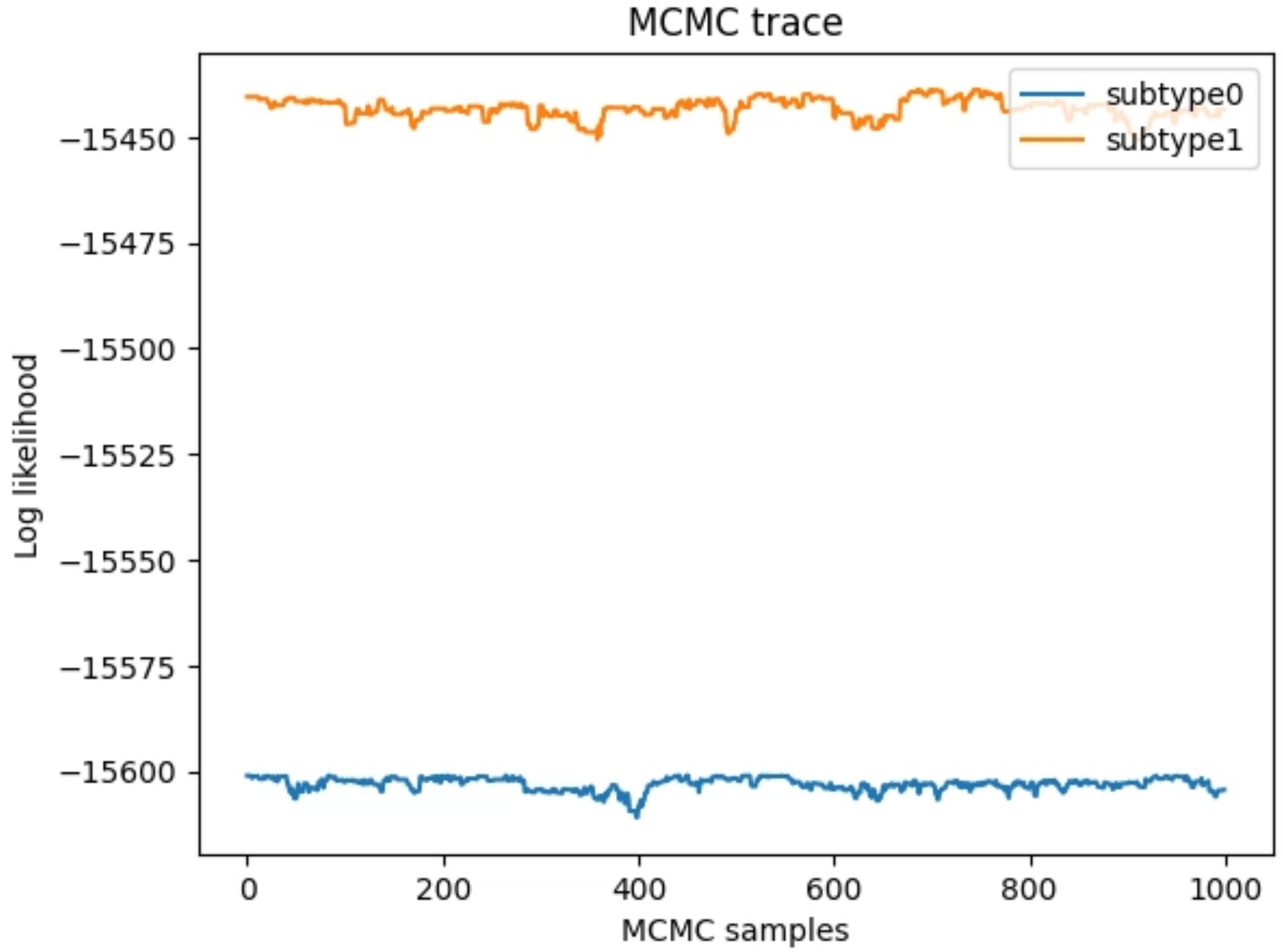
Progress so far

- Worked out mathematical model for genetics-informed likelihood and update equations for genetic weights
- Implemented all changes pySuStaIn
- Preliminary results in Clinical data
- Built synthetic data model

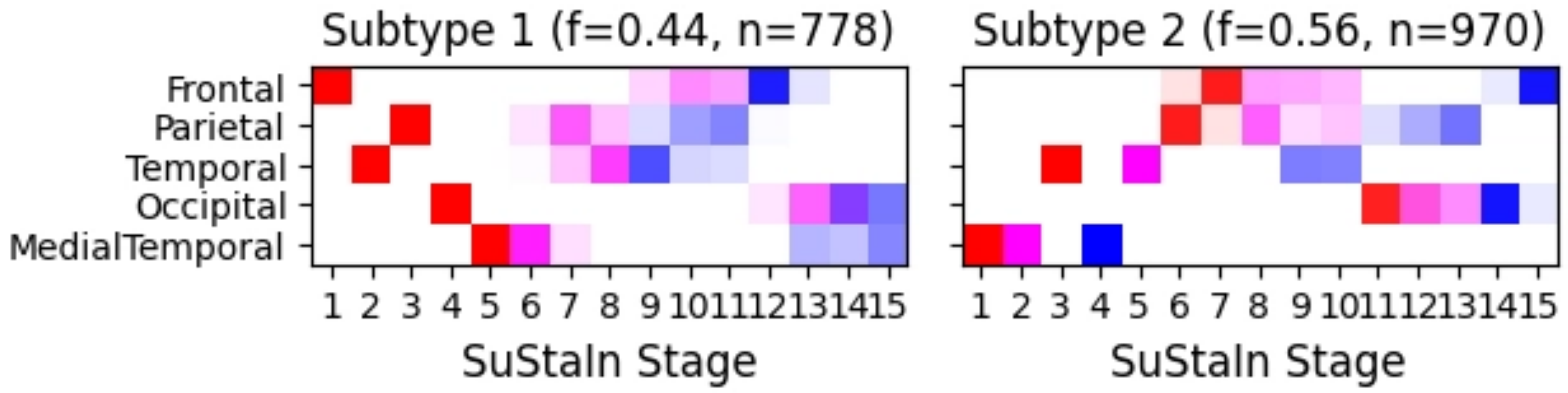
Preliminary results on clinical data

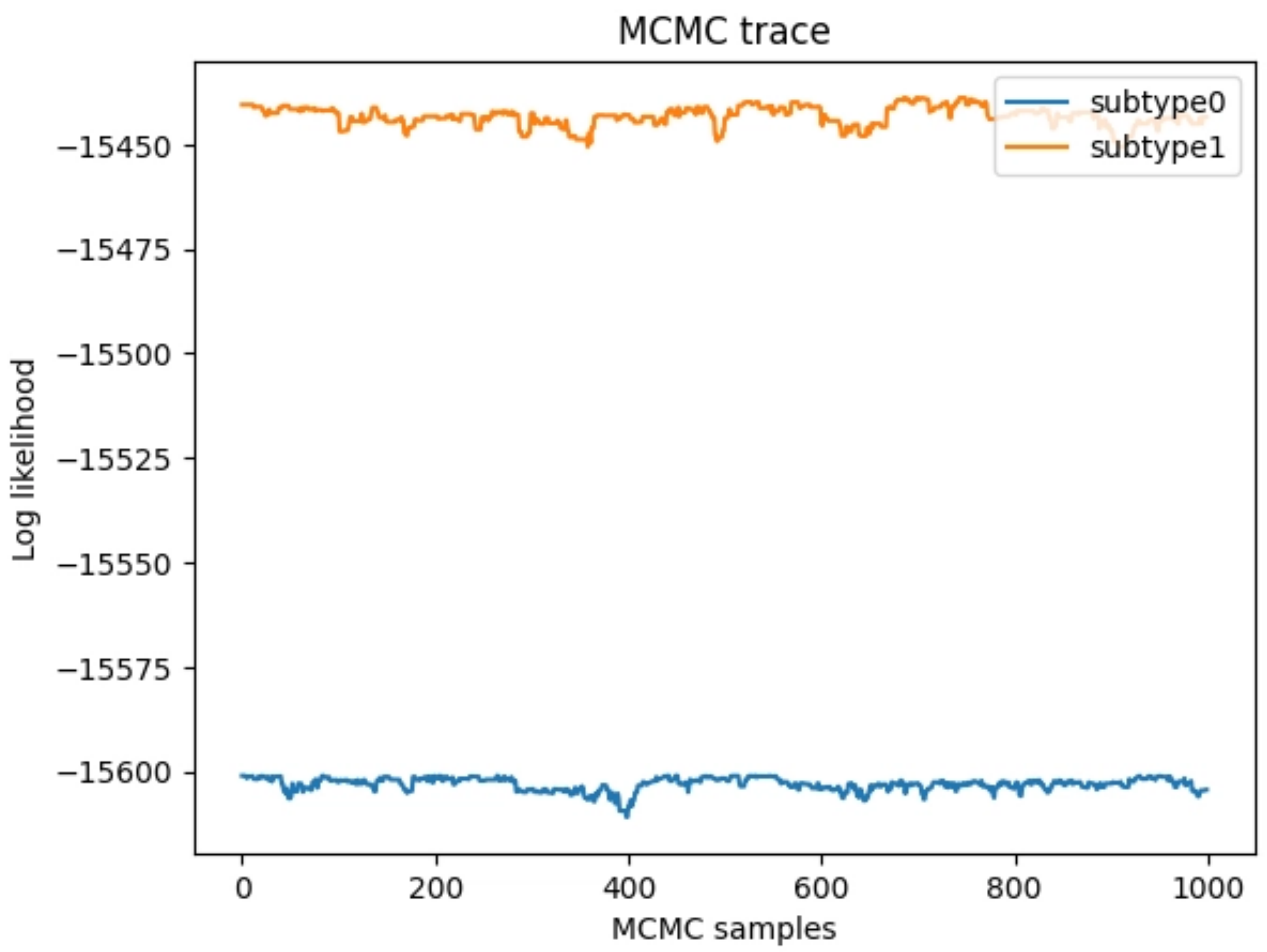
- Ran APOE-SuStaIn on AB+ cohort ADNI
- Estimated genetic weights:
  Subtype 0: [0.6768, 0.2907, 0.0325]
  Subtype 1: [0.4329, 0.4194, 0.1476]
- genetic fractions for whole cohort:
  [0.5423, 0.3617, 0.0960]



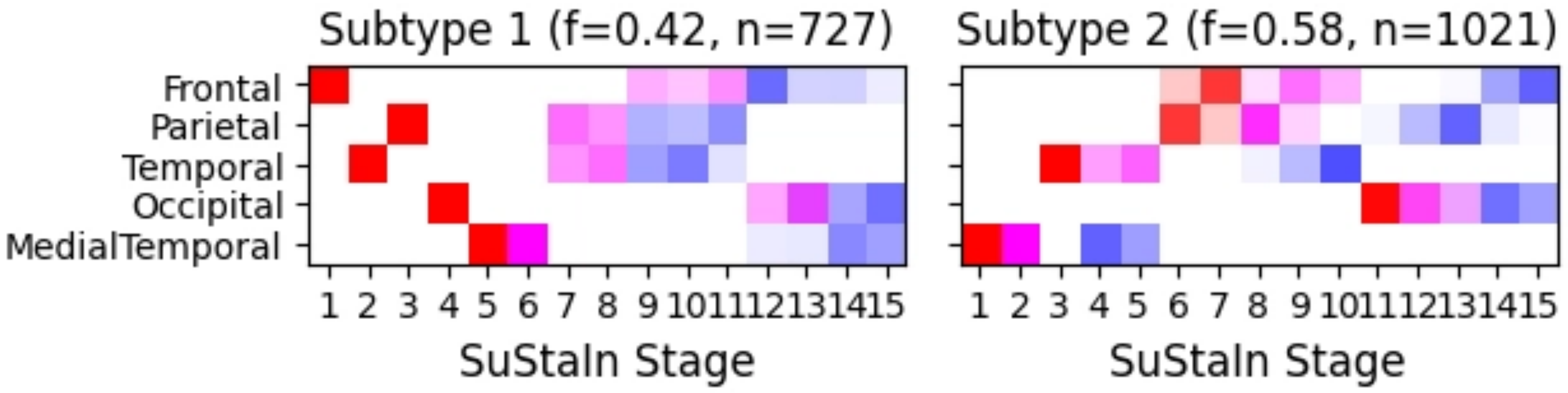


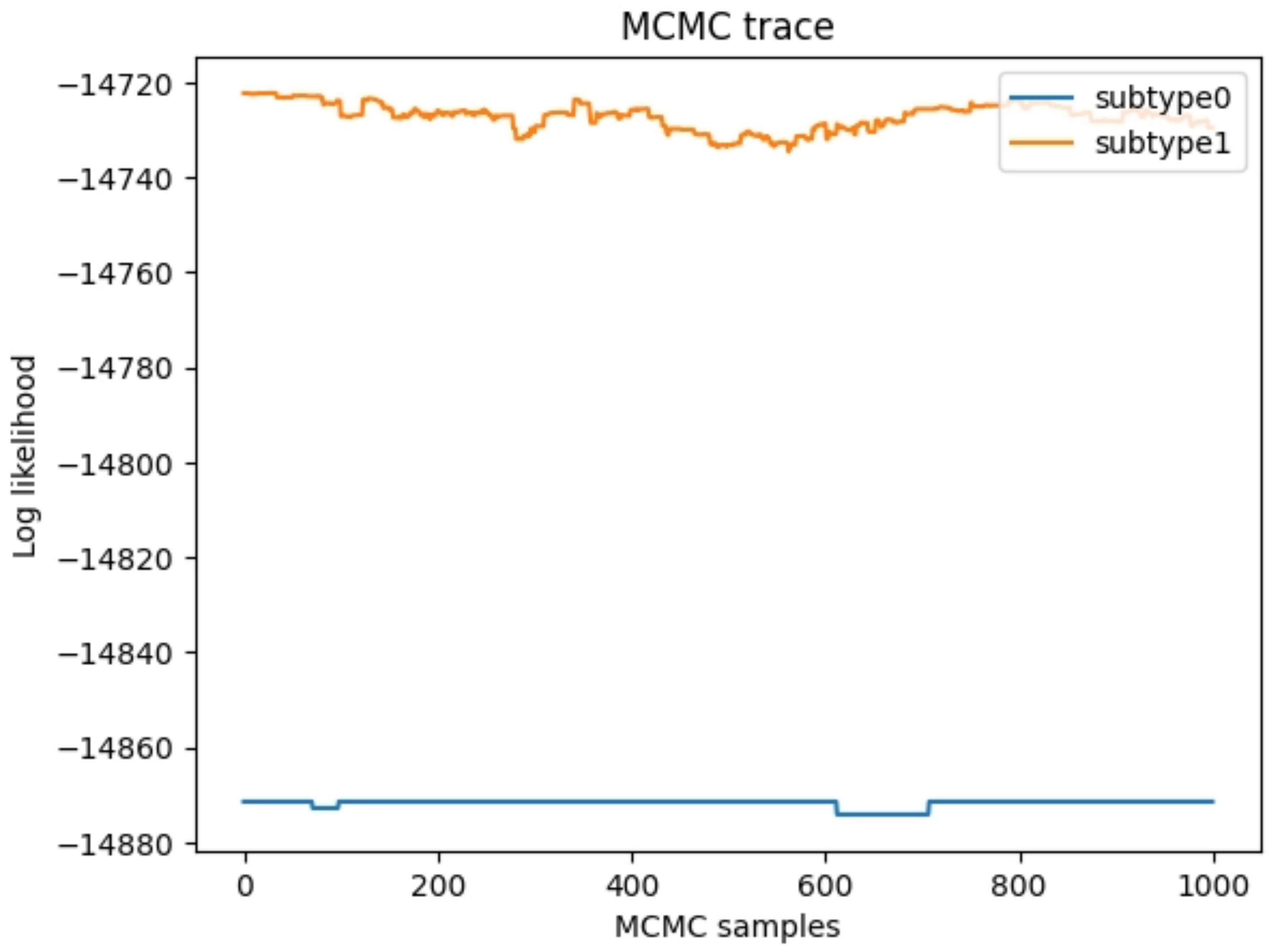
Comparison to baseline SuStaIn





APOE-Informed SuStaIn





Baseline SuStaIn

Simulated data experiments

- APOE status is generated taking in a pre-defined matrix of weights;
- potential scenarios based on genetic influence

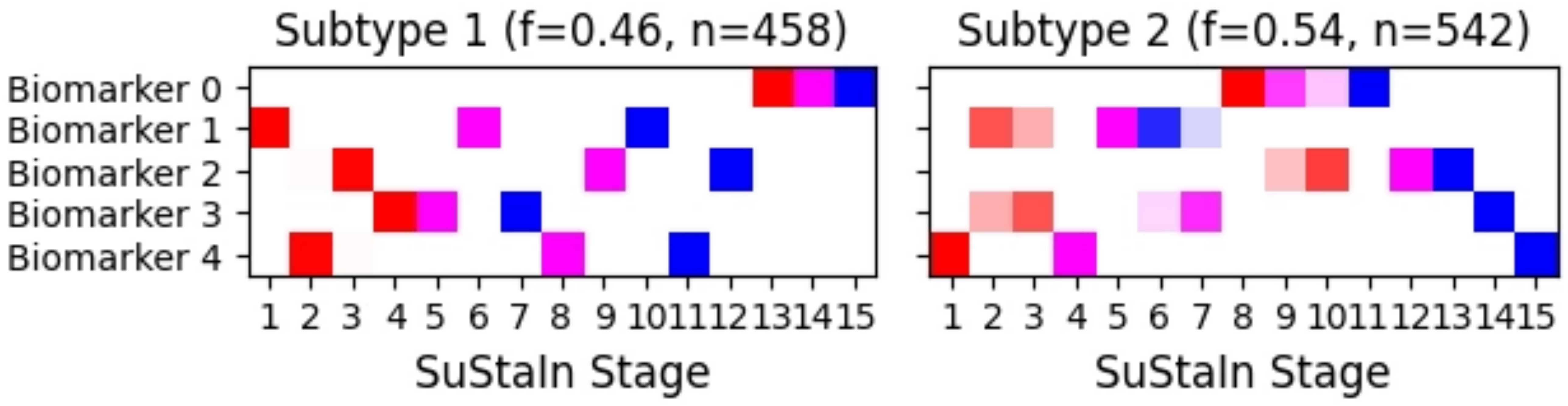
$$W_{Strong} = \begin{bmatrix} 0.80 & 0.15 & 0.05 \\ 0.10 & 0.30 & 0.60 \end{bmatrix}$$

$$W_{Subtle} = \begin{bmatrix} 0.45 & 0.35 & 0.20 \\ 0.30 & 0.40 & 0.30 \end{bmatrix}$$

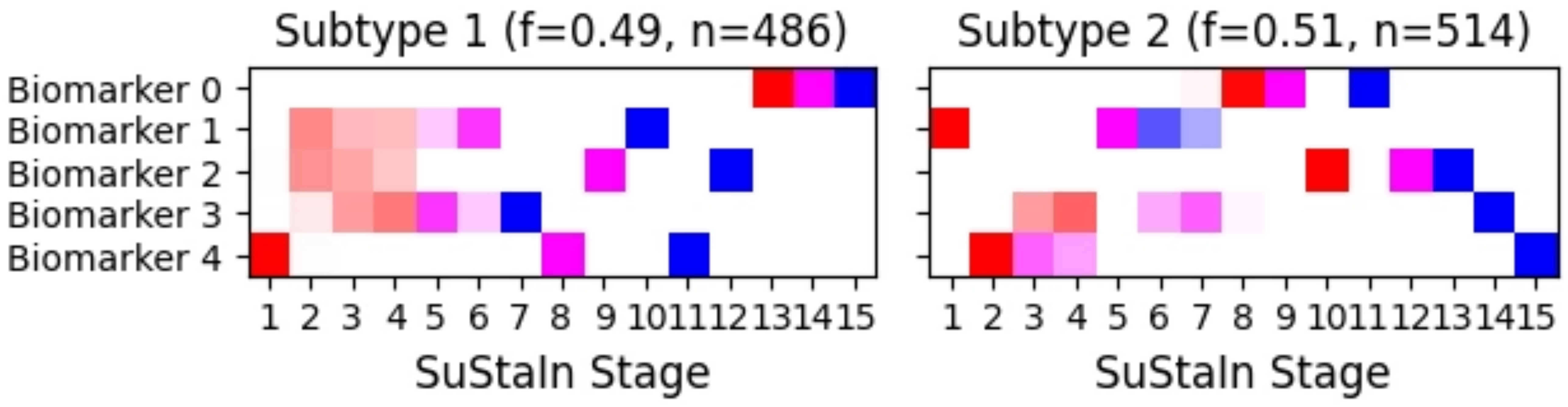
$$W_{Uniform} = \begin{bmatrix} 0.33 & 0.33 & 0.33 \\ 0.33 & 0.33 & 0.33 \end{bmatrix}$$

Preliminary results on simulated data

- Simulated 2 subtypes with distinct sequences and genetic weights:
-
```
W_true_target = np.array([
    [0.80, 0.15, 0.05],  #
    [0.10, 0.30, 0.60]   #
])
```
- Genetics-informed estimated parameters: [Cat_0: 0.1030, Cat_1: 0.3109, Cat_2: 0.5860] [Cat_0: 0.7713, Cat_1: 0.1503, Cat_2: 0.0784]
- Sequences estimated by the extended model are sharper, less uncertainty compared to Baseline Model



Genetics-informed SuStaIn pos var diagram

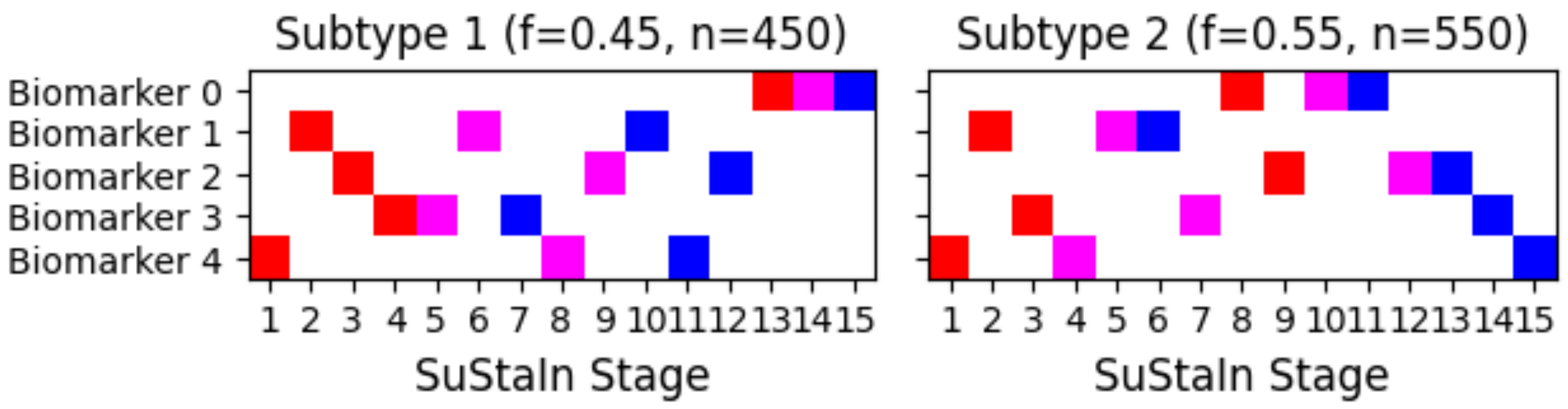


Baseline-SuStaIn pos var diagram

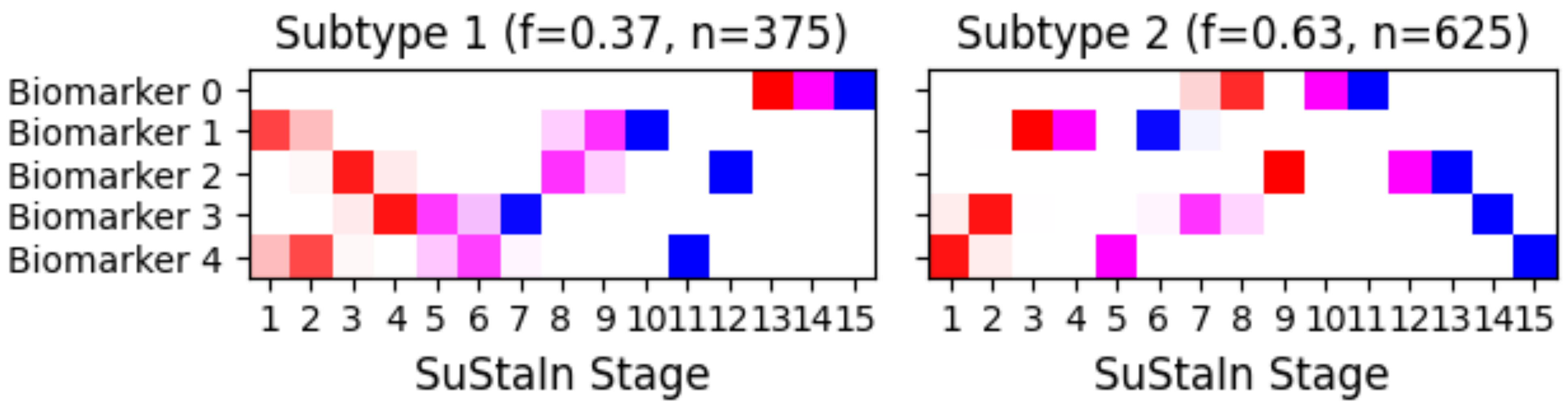
Is it safe? Uninformative prior scenario

- We aim to check what happens if weights are uniform: $W = \begin{bmatrix} 0.33 & 0.33 & 0.33 \\ 0.33 & 0.33 & 0.33 \end{bmatrix}$
- If genetics are uninformative, the model should behave exactly like SuStaIn and not introduce errors;
- Estimated genetic weights:

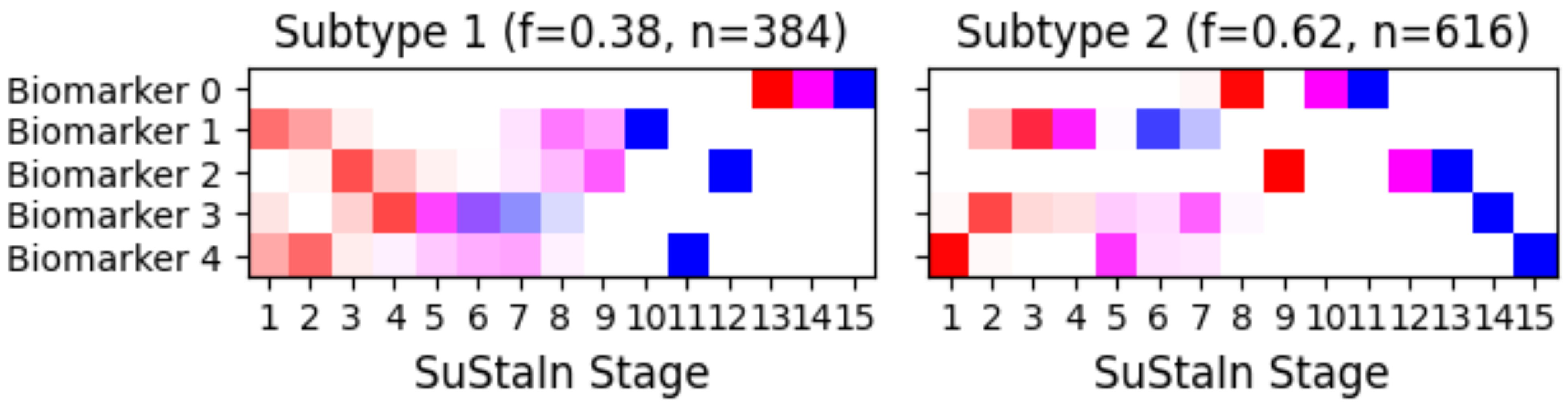
$$W = \begin{bmatrix} 0.33 & 0.33 & 0.33 \\ 0.31 & 0.35 & 0.33 \end{bmatrix}$$



Ground Truth Progression Pattern



APOE-Informed SuStaIn



Baseline SuStaIn

Next steps

- 1. Make last algorithm decisions based on EM likelihood convergence
- 2. Robust synthetic evaluations
- 3. Find the benefit of my model in clinical data: better correlations with clinical metrics, improved MCI to AD predictive power, reduced uncertainty